Calculus II

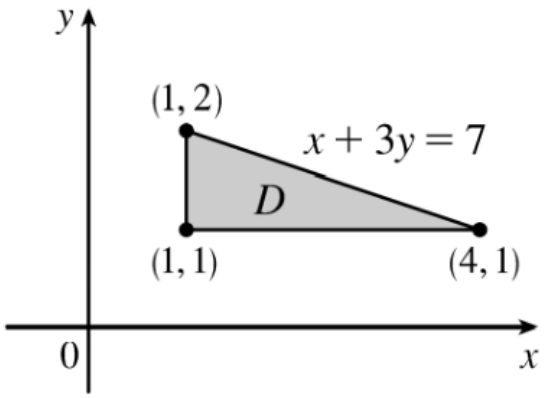
Solution to Assignment 10

Part I. Regular Problems

1.

$$\int_0^1 \int_0^{s^2} \cos\left(s^3\right) dtds = \int_0^1 \left[t\cos\left(s^3\right)\right]_{t=0}^{t=s^2} ds = \int_0^1 s^2\cos\left(s^3\right) ds = \frac{1}{3}\sin\left(s^3\right)\Big|_0^1$$
$$= \frac{1}{3}(\sin 1 - \sin 0) = \frac{1}{3}\sin 1$$

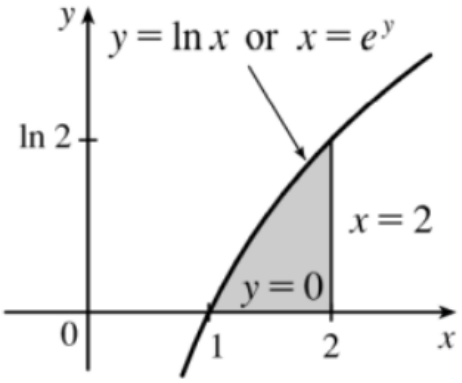
2. The region looks as follows:



$$V = \int_1^2 \int_1^{7-3y} xydxdy = \int_1^2 \left[\frac{1}{2}x^2y\right]_{x=1}^{x=7-3y} dy$$
$$= \frac{1}{2}\int_1^2 y\left[(7-3y)^2 - 1\right] dy = \frac{1}{2}\int_1^2 \left(48y - 42y^2 + 9y^3\right) dy$$
$$= \frac{1}{2}\left[24y^2 - 14y^3 + \frac{9}{4}y^4\right]_1^2 = \frac{31}{8}$$

3. Because the region of integration is

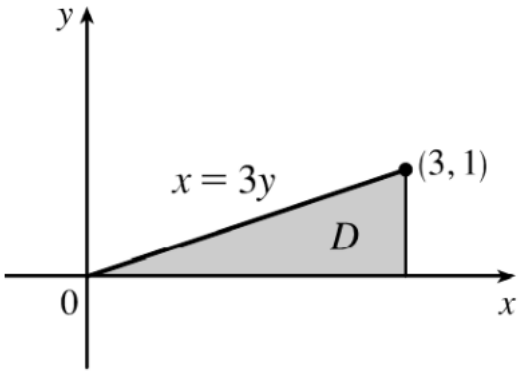
$$D = \{(x, y) \mid 0 \le y \le \ln x, 1 \le x \le 2\} = \{(x, y) \mid e^y \le x \le 2, 0 \le y \le \ln 2\}$$



we have

$$\int_1^2 \int_0^{\ln x} f(x, y) dy dx = \iint_D f(x, y) dA = \int_0^{\ln 2} \int_{e^y}^2 f(x, y) dx dy$$

4. The region looks as follows:



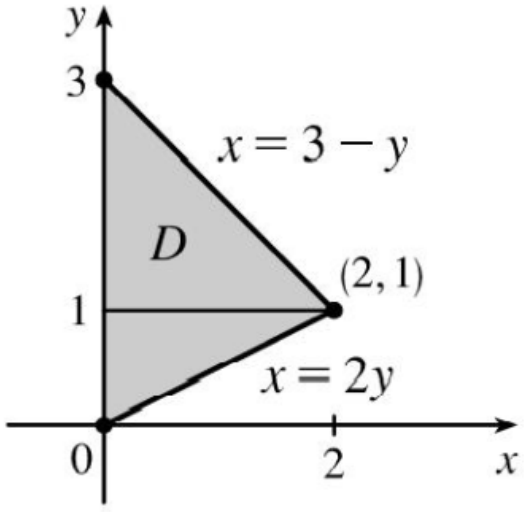
$$\int_0^1 \int_{3y}^3 e^{x^2} dx dy = \int_0^3 \int_0^{x/3} e^{x^2} dy dx = \int_0^3 \left[e^{x^2} y \right]_{y=0}^{y=x/3} dx$$
$$= \int_0^3 \left(\frac{x}{3} \right) e^{x^2} dx = \frac{1}{6} e^{x^2} \Big]_0^3 = \frac{e^9 - 1}{6}$$

5.

$$D = \{(x,y) \mid 0 \le x \le 1, -x+1 \le y \le 1\} \cup \{(x,y) \mid -1 \le x \le 0, x+1 \le y \le 1\}$$
$$\cup \{(x,y) \mid 0 \le x \le 1, -1 \le y \le x-1\} \cup \{(x,y) \mid -1 \le x \le 0, -1 \le y \le -x-1\}, \quad \text{all type I.}$$

$$\iint_D x^2 dA = \int_0^1 \int_{1-x}^1 x^2 dy dx + \int_{-1}^0 \int_{x+1}^1 x^2 dy dx + \int_0^1 \int_{-1}^{x-1} x^2 dy dx + \int_{-1}^0 \int_{-1}^{-x-1} x^2 dy dx$$
$$= 4\int_0^1 \int_{1-x}^1 x^2 dy dx \quad \left[\text{ by symmetry of the regions and because } f(x,y) = x^2 \ge 0\right]$$
$$= 4\int_0^1 x^3 dx = 4\left[\frac{1}{4}x^4\right]_0^1 = 1$$

6. The region looks as follows:

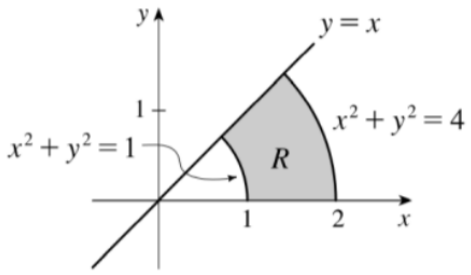


$$\iint_D f(x,y) dA = \int_0^1 \int_0^{2y} f(x,y) dx dy + \int_1^3 \int_0^{3-y} f(x,y) dx dy$$
$$= \int_0^2 \int_{x/2}^{3-x} f(x,y) dy dx$$

7. (a)

$$\iint_D e^{-x^2-y^2} dA = \int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} r dr d\theta = \int_{-\pi/2}^{\pi/2} d\theta \int_0^2 r e^{-r^2} dr$$
$$= [\theta]_{-\pi/2}^{\pi/2} \left[-\frac{1}{2}e^{-r^2}\right]_0^2 = \pi\left(-\frac{1}{2}\right)\left(e^{-4} - e^0\right) = \frac{\pi}{2}\left(1 - e^{-4}\right)$$

(b) The region looks as follows:

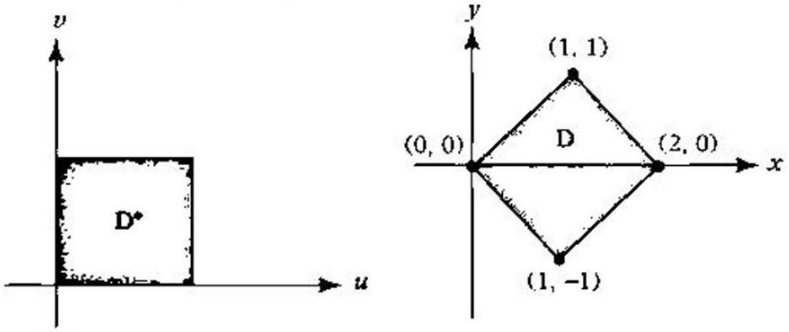


R is the region shown in the figure, and can be described by $R = \{(r,\theta) \mid 0 \le \theta \le \pi/4, 1 \le r \le 2\}$. Thus $\iint_R \arctan(y/x)dA = \int_0^{\pi/4}\int_1^2 \arctan(\tan\theta)rdrd\theta$ since $y/x = \tan\theta$. Also, $\arctan(\tan\theta) = \theta$ for $0 \le \theta \le \pi/4$, so the integral becomes $\int_0^{\pi/4}\int_1^2 \theta r dr d\theta = \int_0^{\pi/4}\theta d\theta \int_1^2 r dr = \left[\frac{1}{2}\theta^2\right]_0^{\pi/4}\left[\frac{1}{2}r^2\right]_1^2 = \frac{\pi^2}{32}\cdot\frac{3}{2} = \frac{3}{64}\pi^2$.

8. The cone $z = \sqrt{x^2+y^2}$ intersects the sphere $x^2+y^2+z^2 = 1$ when $x^2+y^2+\left(\sqrt{x^2+y^2}\right)^2 = 1$ or $x^2+y^2 = \frac{1}{2}$. So

$$V = \iint_{x^2+y^2\le 1/2}\left(\sqrt{1-x^2-y^2}-\sqrt{x^2+y^2}\right)dA = \int_0^{2\pi}\int_0^{1/\sqrt{2}}\left(\sqrt{1-r^2}-r\right)rdrd\theta$$
$$= \int_0^{2\pi}d\theta\int_0^{1/\sqrt{2}}\left(r\sqrt{1-r^2}-r^2\right)dr = [\theta]_0^{2\pi}\left[-\frac{1}{3}\left(1-r^2\right)^{3/2}-\frac{1}{3}r^3\right]_0^{1/\sqrt{2}}$$
$$= 2\pi\left(-\frac{1}{3}\right)\left(\frac{1}{\sqrt{2}}-1\right) = \frac{\pi}{3}(2-\sqrt{2})$$

9. This transformation is linear and takes $(0,0)$ to $(0,0)$, $(1,0)$ to $(1,1)$, $(1,1)$ to $(2,0)$, and $(0,1)$ to $(1,-1)$, so it takes the parallelogram D^* to D, as in the figure.



The integrand is

$$x^2 - y^2 = (u+v)^2 - (u-v)^2 = 4uv$$

and the Jacobian determinant is

$$\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

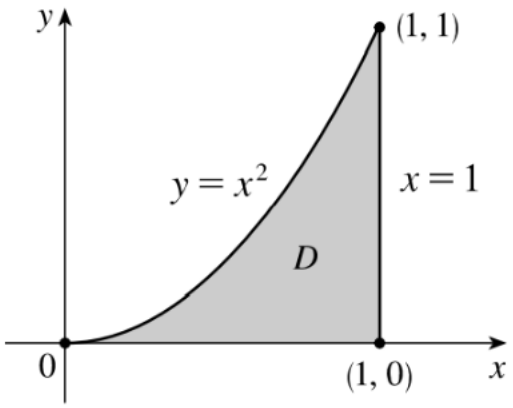
so the change of variables formula gives

$$\iint_D \left(x^2 - y^2\right) dxdy = \iint_D 4uv \cdot |-2| dudv = 8\int_0^1 \int_0^1 uvdudv = 2.$$

Part II. Group Problems

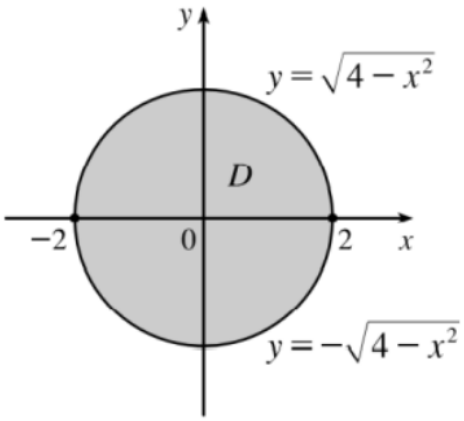
1. (a)
$$\iint_D y\sqrt{x^2-y^2}dA = \int_0^2 \int_0^x y\sqrt{x^2-y^2}dydx = \int_0^2 \left[-\frac{1}{3}\left(x^2-y^2\right)^{3/2}\right]_{y=0}^{y=x} dx$$
$$= \int_0^2 \left[0 + \frac{1}{3}\left(x^2\right)^{3/2}\right] dx = \int_0^2 \frac{1}{3}x^3 dx$$
$$= \frac{1}{3}\cdot\frac{1}{4}\left[x^4\right]_0^2 = \frac{1}{12}(16-0) = \frac{4}{3}$$

(b) The region looks as follows:



$$\int_0^1 \int_0^{x^2} x\cos y dydx = \int_0^1 \left[x\sin y\right]_{y=0}^{y=x^2} dx = \int_0^1 x\sin x^2 dx$$
$$= -\frac{1}{2}\left[\cos x^2\right]_0^1 = -\frac{1}{2}(\cos 1 - \cos 0) = \frac{1}{2}(1-\cos 1)$$

(c) The region looks as follows:



$$\begin{aligned}
&\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}(2x-y)dydx \\
&= \int_{-2}^{2}\left[2xy-\frac{1}{2}y^2\right]_{y=-\sqrt{4-x^2}}^{y=\sqrt{4-x^2}}dx \\
&= \int_{-2}^{2}\left[2x\sqrt{4-x^2}-\frac{1}{2}\left(4-x^2\right)+2x\sqrt{4-x^2}+\frac{1}{2}\left(4-x^2\right)\right]dx \\
&= \int_{-2}^{2}4x\sqrt{4-x^2}dx = -\frac{4}{3}\left[\left(4-x^2\right)^{3/2}\right]_{-2}^{2} = 0
\end{aligned}$$

[Or, note that $4x\sqrt{4-x^2}$ is an odd function, so $\int_{-2}^{2}4x\sqrt{4-x^2}dx = 0$.]